

# PENETRATION TESTING REPORT

## FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

|                                                        |                                                                                                                                |
|--------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>Pentester Name<br/>(Cybersecurity Professional)</b> | Emmanuel John                                                                                                                  |
| <b>Program/Batch</b>                                   | B082-Networkwalks                                                                                                              |
| <b>Date</b>                                            | 19 August 2026                                                                                                                 |
| <b>Modules completed</b>                               | W2-PM1 (Maltego)<br>W2-PM5 (Zenmap Scanning)                                                                                   |
| <b>Client/Target</b>                                   | 1. Networkwalks (secured written permission already)<br>2. My own local LAN Network                                            |
| <b>Permission secured from client?</b>                 | Yes                                                                                                                            |
| <b>Phases covered</b>                                  | <b>Phase 1:</b> Reconnaissance & Footprinting<br><b>Phase 2:</b> Scanning & Network Discovery<br><b>Phase 3-5:</b> In Progress |

### 1. Liability Disclaimer

I have performed these activities only on the systems & devices where I had secured written permission or the devices/systems that I own myself. All these materials are for education and research purpose only. Do not use anything from here to break the law. The instructor, the authors and Networkwalks are not responsible for what you do with this knowledge. Every action you take is your own responsibility. Misuse can lead to criminal charges, heavy fines, loss of your job and a permanent record. In most countries unauthorised access is a crime even when nothing is damaged.

## 2. Introduction

This report covers footprinting the networkwalks.com domain using multiple Kali Linux tools (W2-PM1) and scanning my own local network with Zenmap (W2-PM5). One module covers the footprinting phase and the other covers the scanning phase, so together they show how an attacker moves from gathering public information to mapping live hosts on a network. It is the Week 2 part of my ongoing internship program at Networkwalks.

All commands were run in Kali Linux (footprinting) and on a Windows PC with Zenmap installed (scanning). Every step below includes the exact command used, the result I observed, a screenshot as evidence, and a short note on why the finding matters from an attacker's point of view.

## 3. Tools Used

The table below lists each tool used in this report and its purpose.

| Tool              | Purpose                                                          |
|-------------------|------------------------------------------------------------------|
| Windows           | Operating systems used for reconnaissance activities             |
| Zenmap (Nmap GUI) | Scan the local subnet to find live hosts, IPs and MAC addresses. |
| windows cmd       | Local IP and MAC address identification                          |

## 4. Activities Performed

### 4.1 Footprinting & Reconnaissance with Maltego

I performed reconnaissance of the **networkwalks.com** domain using **Maltego** as part of the W2-PM1 practical activity.

I entered the target domain into Maltego and used the available transforms to gather publicly available information related to the target. Maltego displayed the discovered information as connected entities in a graphical format.

During the reconnaissance process, **Maltego identified an email address associated with the target domain**. This demonstrated how publicly available information can be collected and connected during the footprinting phase.

The discovered email information can be useful from a cybersecurity perspective because publicly exposed email addresses may provide additional information about an organization's online presence and can become a starting point for further **authorized security assessment**.

**Result:** Maltego successfully generated a graphical footprint of the target and identified an **email address associated with the target domain**.

## 4.2 Network Scanning with Zenmap

For the second activity, I used **Zenmap (Nmap GUI)** to perform network discovery on my own local network.

First, I identified my network range and entered the subnet **192.168.56.0/24** into Zenmap. I selected the **Ping Scan** profile to identify active hosts without performing a port scan.

The command executed by Zenmap was:

```
nmap -sn 192.168.**.0/24
```

The scan checked **256 IP addresses** and identified **1 live host**. The discovered host was:

**IP Address: 192.168.56.1**

The Nmap output showed that the host was **up**. The complete scan was finished in approximately **11.82 seconds**.

**Result:** One active host, **192.168.56.1**, was identified in the scanned **192.168.56.0/24** network.

### Zenmap Scan Observation

| Parameter            | Result                      |
|----------------------|-----------------------------|
| Target Network       | 192.168.56.0/24             |
| Scan Type            | Ping Scan                   |
| Command              | nmap -sn<br>192.168.56.0/24 |
| IP Addresses Scanned | 256                         |
| Live Hosts           | 1                           |
| Discovered Host      | 192.168.56.1                |
| Host Status          | Up                          |
| Scan Duration        | 11.82 seconds               |

## 5. Risk Analysis / Impact

Based on the information collected during the footprinting and network scanning activities, I identified the following potential risks.

| # | Risk / Finding                      | Evidence / Observation                                                                         | Potential Impact                                                                                                                          | Risk Level |
|---|-------------------------------------|------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|------------|
| 1 | Email address exposed               | Maltego identified an email address associated with the target domain.                         | Publicly available email information may assist further reconnaissance or targeted phishing attempts.                                     | ● Medium   |
| 2 | Live host detected on local network | Zenmap identified <b>192.168.56.1</b> as an active host in the <b>192.168.**.0/24</b> network. | Network discovery can help identify active devices and should be regularly reviewed to ensure that only authorized devices are connected. | ● Medium   |

**Risk level key:** ● Critical ● Medium ● Low

The risks above are observations from the footprinting and scanning exercises, not confirmed vulnerabilities.

The practical exercises primarily involved information gathering and host discovery. No exploitation or vulnerability validation was performed as part of these two modules.

Therefore, the presence of information such as a software version, IP address or DNS record does not by itself mean that the system is vulnerable. Further authorized security testing would be required to confirm any actual vulnerability.

## 6. Recommendations

Based on the observations from these activities, I recommend the following security improvements:

1. **Review publicly exposed technology information**

Organizations should regularly review what information about their web technologies, CMS and plugins is publicly visible.

2. **Keep software updated**

CMS platforms, plugins and other web technologies should be regularly updated and reviewed against current security advisories.

3. **Review HTTP headers**

HTTP response headers should be reviewed to determine whether unnecessary technical information is being exposed.

4. **Review DNS records regularly**  
DNS records should be checked periodically to ensure that only required information and services are publicly exposed.
5. **Properly configure and monitor the WAF**  
Keep the WAF (ModSecurity) enabled and tuned, since it already blocks naive attacks.
6. **Perform regular internal network discovery**  
Organizations should periodically scan their own networks to identify active devices.
7. **Investigate unknown devices**  
Any unexpected device discovered during network scanning should be investigated and verified.
8. **Maintain network documentation**  
Network topology and device information should be documented and updated regularly.
9. **Perform security testing with authorization**  
Reconnaissance and scanning should only be performed against systems and networks where appropriate authorization has been provided.

## 7. Conclusion

During Week 2 of my Cybersecurity & Ethical Hacking internship, I completed practical activities related to **footprinting, reconnaissance and network scanning** using Maltego and Zenmap.

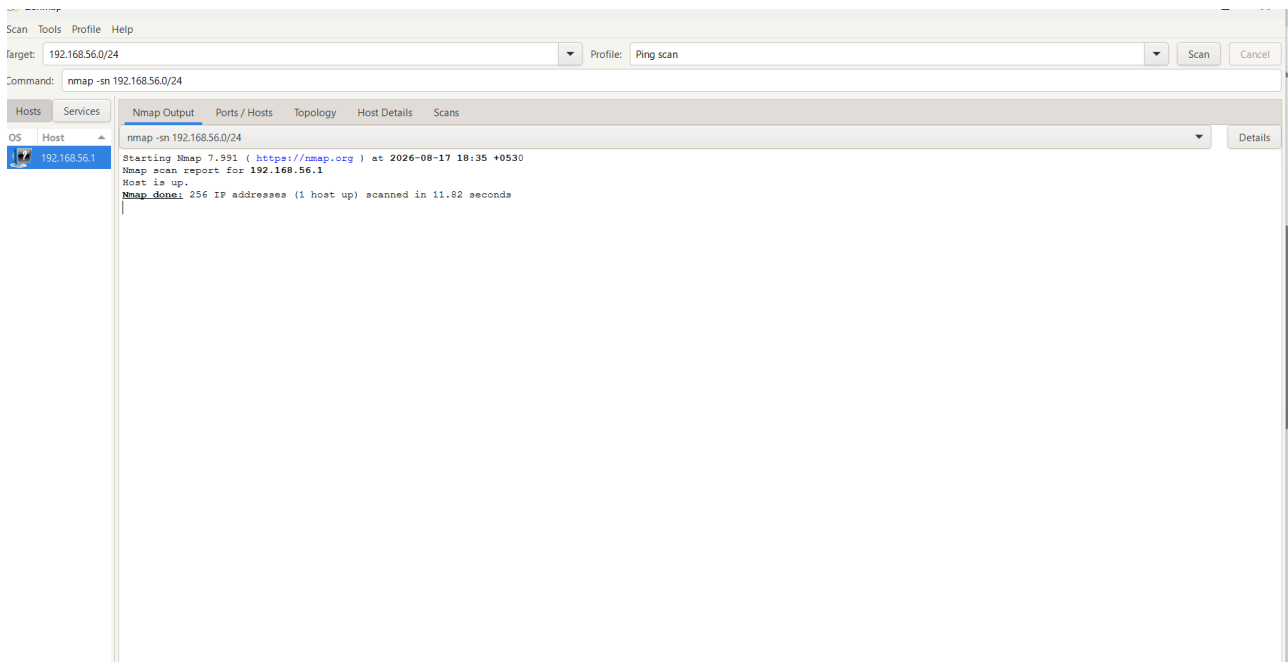
In the footprinting activity, I used **Maltego** to perform graphical reconnaissance of the target domain. I learned how publicly available information can be represented as connected entities and relationships, making it easier to understand the target's footprint.

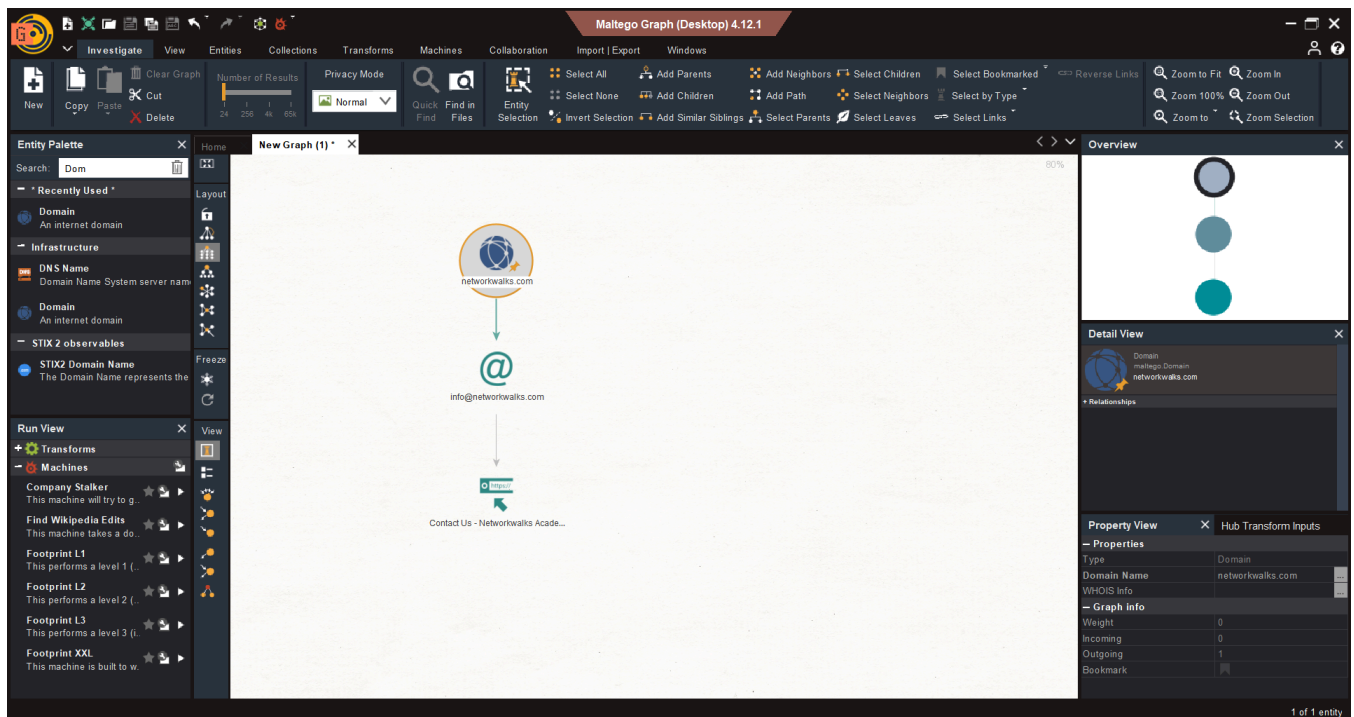
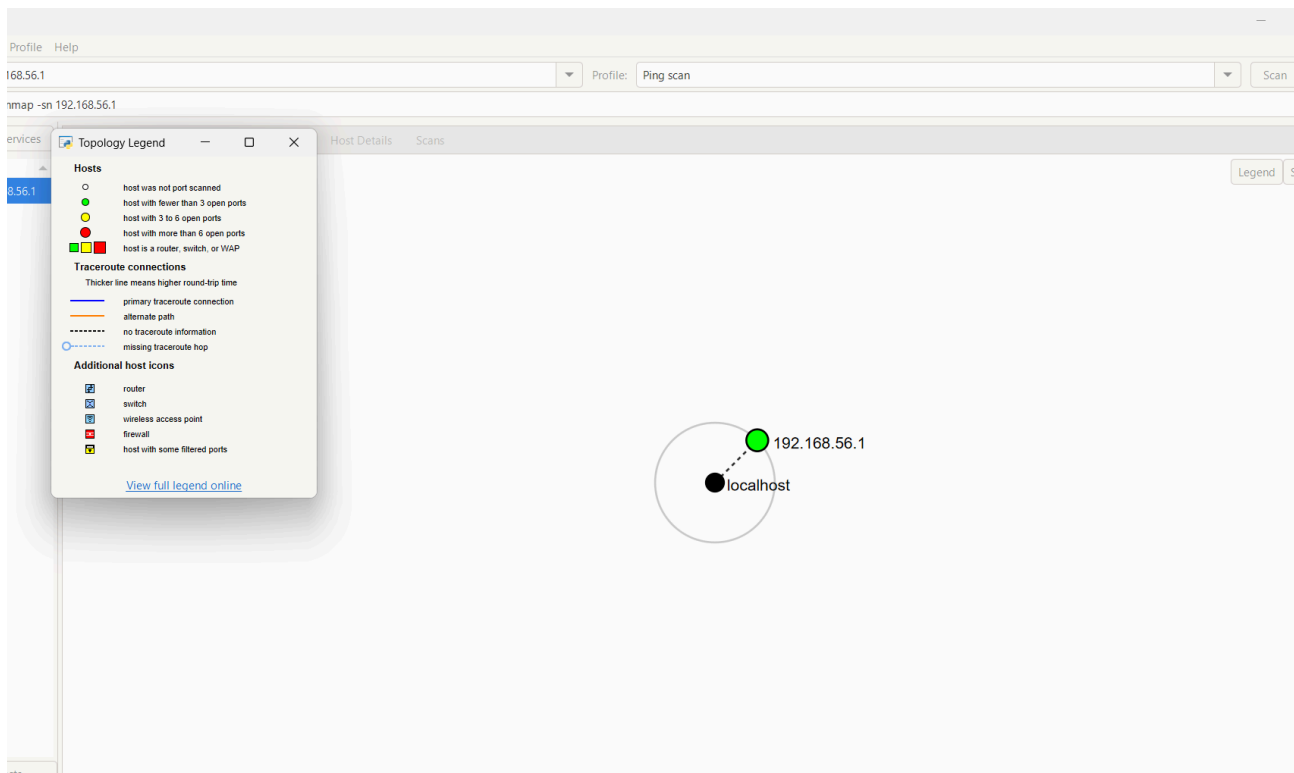
In the network scanning activity, I used **Zenmap** to scan my own local network. The scan identified **1 live host** and provided available IP and MAC address information. I also used the Topology feature to visualize the network.

These activities helped me understand that reconnaissance and network scanning are important parts of cybersecurity. They allow security professionals to gather information about an environment before performing further authorized security testing.

I also learned that reconnaissance and scanning must always be performed within an **authorized scope**.

## 8. Evidences Collected





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### Project Information

**Program Name:** Cybersecurity program at Networkwalks | **Week:** 02 | **Repository:** GitHub